RUNGTA INTERNATIONAL SKILLS UNIVERSITY

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SCHOOL OF INFORMATION TECHNOLOGY

BACHELOR OF COMPUTER APPLICATION

<table>
<tr><td>PS No.</td><td>7</td></tr>
<tr><td>Project Title</td><td>Shopping Cart System</td></tr>
<tr><td>Problem Statement</td><td>Develop a console shopping cart where the user can view products, add or remove items and generate a final bill with discount and GST.
Language: Python 3 Concepts: Lists, Dictionaries, Functions, Billing
Note: Products: Pen Rs.10, Notebook Rs.50, Bag Rs.500. GST 18%. 10% discount if subtotal is more than 500.</td></tr>
<tr><td>Test Cases / Case Statements (4 cases)</td><td>
<table>
<tr><th>INPUT</th><th>EXPECTED OUTPUT</th></tr>
<tr><td colspan="2">Test Case 1: Normal Input</td></tr>
<tr><td>Add to cart: Notebook, Quantity: 2</td><td>Notebook x2 added | Cart subtotal: 100</td></tr>
<tr><td colspan="2">Test Case 2: Boundary / Minimum Input</td></tr>
<tr><td>Add to cart: Bag, Quantity: 1
Checkout</td><td>Subtotal: 500 | Discount: 0 | GST: 90
Grand Total: 590</td></tr>
<tr><td colspan="2">Test Case 3: Invalid Input</td></tr>
<tr><td>(a) Add to cart: Laptop
(b) Add to cart: Pen, Quantity: -1</td><td>(a) Invalid product. Available: Pen, Notebook, Bag
(b) Quantity must be at least 1.</td></tr>
<tr><td colspan="2">Test Case 4: Different Valid Scenario</td></tr>
<tr><td>(a) Add Bag x2
(b) Add Pen x5
(c) Remove Pen
(d) Checkout</td><td>(c) Pen removed. Cart: Bag x2 = 1000
(d) Subtotal: 1000 | Discount: 100 | GST: 162
Grand Total: 1062</td></tr>
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</td></tr>
<tr><td>Bare Minimum Requirements</td><td>1. Write the solution in Python 3 (one .py file) and run it from the console.
2. Keep the product catalogue in a dictionary.
3. Add, remove and view cart items using functions.
4. Calculate subtotal, discount, GST and grand total.
5. Validate the product name and quantity.</td></tr>
<tr><td>Instruction</td><td>1. Understand the problem statement and define the problem clearly.
2. Develop the solution using proper logic and programming concepts.
3. Implement all required features and pass the given test cases.
4. Test and debug the project to ensure correct output.
5. Prepare the documentation/report and upload the code and report to GitHub.
6. Complete and submit the project within the given deadline.</td></tr>
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